CoEP: Amortizing Recurring Repairs in Programming Assignments with LLM-Synthesized Repair Tools

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We present CoEP, a cost-effective approach that repairs students' programming assignments by having a large language model (LLM) synthesize an assignment-specific repair tool rather than a patch for every submission. Recently, LLM-based program repair has substantially improved the share of student submissions that can be repaired automatically. However, these techniques query the model once per buggy submission, so their spending grows with enrollment and recurs every term, even though many submissions need "recurring repairs"—edits that other submissions to the same assignment also need. This paper aims to amortize recurring repairs at the level of the assignment without sacrificing coverage. One major challenge is that recurring repairs rarely recur verbatim, making it difficult to generalize a repair beyond the program it came from. To address this, CoEP selects worked pairs by clustering the class's edit scripts, synthesizes misconception-named rewrite rules validated against the assignment's test cases, amplifies those test cases by differential testing against the class's correct programs, repairs every submission locally with the rules and a hunk-level reference transplant, and learns new rules from the few submissions it escalates to a per-submission query. Every repair CoEP returns passes the amplified tests, whatever the LLM writes. Experimental results on 2,970 buggy programs from 11 assignments show that CoEP repairs 99.4% of them, compared with 98.9% for per-submission LLM repair and 85.9% for reference-guided repair, while producing 33% smaller patches and spending 21.3 and 27.3 times less on two LLMs; on a first-year course, 99.4% of its repairs also pass held-out test cases, against 97.9% of per-submission repairs.

CCS Concepts: • **Software and its engineering** → **Software testing and debugging**; *General programming languages*; • **Social and professional topics** → *Computing education.*

Additional Key Words and Phrases: automated program repair, programming assignments, large language models, program synthesis

1 Introduction

Automated program repair (APR) for programming assignments [1, 9, 11, 36, 44] automatically fixes students' incorrect submissions, so that every student in a course receives immediate and personalized feedback [42]. Given an assignment's problem description, its test cases, and a buggy submission, the technique produces a fixed program that passes every test case, from which a hint or a patch is presented to the student. In practice, hundreds of students attempt the same assignment in a course, and online judges accumulate thousands of submissions to the same problem [24]. In particular, the submissions that already pass every test case form a pool of reference programs on which repair can draw.

Two lines of techniques have been developed for this task. Reference-guided techniques [1, 9, 11, 36] align a buggy program with a structurally matching reference program and transplant the regions in which they differ; they need no model and repair each submission at no marginal cost. LLM-based techniques [7, 14, 22, 23, 38, 40, 41, 44] instead query a large language model (LLM) with the buggy program, which removes the need for a matching reference and extends repair to harder problems. As a result, the share of submissions that can be repaired automatically has seen marked improvement [22, 44].

However, our observation is that per-submission repair re-derives the same repairs over and over. In this work, we define a repair as *recurring* if another buggy submission to the same assignment needs the identical edit to reach its nearest reference program. As shown in Section 3.2, 36.6% of 2,970 buggy programs from 11 assignments need a recurring repair, and 49.9% on first-year course assignments; this is a lower bound, since two submissions that share one misconception rarely share an entire edit. Yet the LLM-based repairer PyDex [44] pays one query for every submission,

$292.77 over these programs on `claude-sonnet-5`, and prompt caching removes only 10.2% of that spend. Reference-guided repair avoids the spend, but Refactory [11] repairs only 54.5% of the buggy programs on online-judge problems rated 1200 and above, where the pool seldom holds a structurally matching program. Consequently, a course must choose between coverage and a model spend that grows with enrollment and recurs every term.

In this paper, we present CoEP (Cost-Effective Program repair), which amortizes recurring repairs by having an LLM synthesize an assignment-specific repair tool once per assignment, rather than a patch once per submission. Our starting point is the observation of Singh et al. [31] that a small *error model*—a set of rewrite rules capturing the mistakes a class makes—corrects most incorrect attempts at an assignment; their error models, however, are written by hand for each assignment. The key insight of CoEP is that the error model can itself be synthesized: a recurring repair is a transformation, a transformation can be written down once as an abstract syntax tree (AST) rewrite rule named after the misconception it repairs, and a builder LLM can write such rules from a handful of worked (buggy, reference) pairs. The deployed tool then repairs most submissions with no model at all, and only what it cannot repair is sent to a conventional per-submission query.

However, a key challenge is that recurring repairs rarely recur verbatim. As reported in Section 3.2, the 2,970 buggy programs need 2,228 distinct edit scripts. Even on cAPS lOCK, where a single rule repairs 108 of the 560 buggy programs, the largest group of programs sharing an identical edit script holds only 31. Thus, a builder must generalize each repair beyond the program it was shown, and the code it writes must be trusted without trusting its own claims about that code. A second obstacle is that the instructor's tests are thin, 7.4 per assignment on the first-year course, so a candidate that passes them may still be wrong, and a tool that searches many candidates is precisely the repairer most likely to find one that fits the tests rather than the problem.

To address these challenges, CoEP treats the LLM as an untrusted synthesizer and validates everything it produces by execution. It performs four stages: (1) *Select*, which pairs each buggy program with its minimum-edit reference program and clusters the class by the resulting edit scripts, choosing one worked pair per cluster; (2) *Synthesize*, a counterexample-guided loop in which the builder writes misconception-named rewrite rules and is shown the worked pairs they still fail to repair, and in which it also writes an input generator whose inputs the class's correct programs label by majority vote, amplifying the instructor's tests T into a suite T^+; (3) *Repair*, which repairs every submission by a breadth-first search over compositions of the frozen rules and a hunk-level reference transplant, escalating to a per-submission query only on abstentions and oversized patches; and (4) *Learn*, which distills the escalated repairs into new rules. We prove that every repair CoEP returns passes T^+ whatever the builder writes, that the rule search returns a shallowest composition, that escalation never loses coverage or enlarges a patch, and that repair terminates (Section 4.7).

Experimental results demonstrate that CoEP markedly reduces the cost of LLM-based repair without sacrificing coverage. We evaluated CoEP with two builders, `claude-sonnet-5` and `gpt-5.4-mini`, on 2,970 buggy programs from 11 assignments, five from a first-year programming course [11] and six Codeforces problems rated 800 to 2000 [24], against the reference-guided Refactory [11] and the LLM-based PyDex [44]. In terms of repair rate, CoEP repaired 99.4% of the programs, where Refactory repaired 85.9% and PyDex 98.9%, and CoEP's frozen tool alone repaired 98.4% without a model call. Regarding patch size, CoEP's patches were 33% smaller than PyDex's. Regarding cost, CoEP spent $13.74 and $1.04 over the benchmark on the two models, 21.3 and 27.3 times less than PyDex, and repaid its one-off build after 11 and 7 submissions per assignment. Regarding correctness beyond the instructor's tests, 99.4% of CoEP's repairs on the first-year course also passed 100 held-out test cases per assignment, against 97.9% of PyDex's. We further demonstrate the efficacy of each stage through ablation, and the generality of CoEP on later

```
1  s=input()
2  c=s[0]
3  if c.islower():
4      if s[1:].isupper():
5          rs=s.swapcase()
6          print(rs)
7      else:
8          print(s)
9  elif s.isupper():
10     es=s.swapcase()
11     print(es)
12 else:
13     print(s)
```

(a) A buggy submission. The rule rewrites line 4 to `if s[1:].upper() == s[1:]`: (RPS = 0.118).

```
def _rule_slice_case_to_equality(source):
    """slice.isupper() fails on an empty slice; compare instead."""
    tree = ast.parse(source)
    out = []
    for node in ast.walk(tree):
        if (isinstance(node, ast.Call) and not node.args
                and not node.keywords
                and isinstance(node.func, ast.Attribute)
                and node.func.attr in ("isupper", "islower")
                and isinstance(node.func.value, ast.Subscript)):
            vs, ve = _span(source, node.func.value)
            value_text = source[vs:ve]
            new_text = "%s.upper() == %s" % (value_text, value_text)
            candidate = _splice(source, node, new_text)
            if candidate != source:
                out.append(candidate)
    return out
```

(b) The rule `slice_case_check_to_equality` written by `claude-sonnet-5` (abridged).

Fig. 1. A recurring misconception on cAPS lOCK (Codeforces 131A) and the synthesized rule that repairs it.

submissions: a tool built from the first 10% of a class repaired 99.4% and 98.2% of the last quarter's buggy programs on the two corpora.

Contributions. We summarize our contributions below:

- **Formulation:** We formulate repair for programming assignments as the synthesis of an assignment-specific repair tool—a set of AST rewrite rules together with an amplified acceptance suite—with an LLM as an untrusted synthesizer, generalizing the hand-written error models of prior work [31] (Sections 3 and 4).
- **Technique and guarantees:** We present CoEP, an algorithm that selects worked pairs by clustering edit scripts, synthesizes misconception-named rules in a counterexample-guided loop, amplifies the tests by differential testing, repairs every submission locally by searching rule compositions and transplanting reference hunks behind a gated fallback, and learns new rules from escalated repairs; we prove its soundness relative to T^+ and three further properties (Section 4).
- **Evaluation:** We evaluate CoEP against reference-guided and LLM-based repair on 2,970 buggy programs from 11 assignments, including the prevalence of recurring repairs, held-out test cases, and splits by submission order. We make CoEP, the 22 synthesized tools, the 10 input generators, every prompt, and every per-submission log publicly available (Data-Availability Statement).

2 Overview

We illustrate CoEP on two assignments of our benchmark, cAPS lOCK and sequential search.

A recurring misconception. cAPS lOCK (Codeforces 131A) asks whether a word was typed with the caps lock on, i.e., whether all its letters, or all but the first, are uppercase. Suppose a student tests the tail of the word with `s[1:].isupper()` at line 4 of Figure 1-(a). The program then fails on a one-letter word such as `"z"`, since the tail is empty and an empty string is not uppercase in Python. 108 of the 560 buggy programs submitted to this problem share this misconception. Per-submission LLM repair queries the model 108 times to rediscover the same fix, and a reference transplant must import a length guard together with the structure around it. Nor do the 108 programs share one edit: they differ elsewhere, so their edits to the nearest correct program differ too.

Synthesizing the tool. CoEP builds the tool before the first repair. Select pairs each of the 560 buggy programs with the correct program it is fewest edits from and clusters the programs by these

```
def search(x, seq):
  for i in range(1, len(seq)+1):
    if x < seq[i-1]:
      return i-1
    elif seq[i-1]<x<seq[i]:
      return i
  return i
```

(a) A buggy submission.

```
def search(x, seq):
  for i in range(len(seq)-1):
    if seq[i] < x <= seq[i+1]:
      return i+1
    elif x <= seq[0]:
      return 0
  return len(seq)
```

(b) Accepted by T; `search(-15, [-6])` returns 1 (RPS = 0.47).

```
def search(x, seq):
  if not seq:
    return 0
  elif x <= seq[0]:
    return 0
  else:
    for i in range(len(seq)-1):
      if seq[i] < x <= seq[i+1]:
        return i+1
    return len(seq)
```

(c) Accepted by T^+ (RPS = 0.84).

Fig. 2. Patches for one submission to sequential search (R1) under the instructor's test cases T and the amplified suite T^+.

edit scripts, yielding 353 worked pairs. Synthesize shows them to a builder LLM, which answers with a Python module of rewrite rules; CoEP runs the rules on the worked pairs and returns the pairs they still fail to repair, for three rounds. On cAPS lOCK, `claude-sonnet-5` wrote 10 rules for $1.23. Figure 1-(b) shows one, `slice_case_check_to_equality`: it locates a case test on a slice and splices in an equality that also holds for the empty slice. The rule was written from one worked pair but is stated over the AST, so it applies wherever the pattern occurs.

Repairing locally. The frozen tool repairs each submission by a breadth-first search over compositions of its rules, validating every candidate by running the tests, and transplants line hunks from the reference programs when no composition passes. On cAPS lOCK, `slice_case_check_to_equality` repaired 108 programs, the tool's rules 156 in all, and the transplant the remaining 404, without a model call; the repair of Figure 1-(a) changes one line, at a relative patch size (RPS) of 0.118.

When the tests are thin. Sequential search (R1) asks for the position at which x would be inserted into a sorted sequence, and its instructor ships 11 test cases, none with a one-element sequence. For the buggy program of Figure 2-(a), the transplant finds the patch of Figure 2-(b), which passes all 11 tests but returns 1 for `search(-15, [-6])`: its guard `x <= seq[0]` sits inside a loop that never runs on a one-element sequence. The builder's input generator draws sorted sequences, one-element ones included, and the class's correct programs label each input by majority. 100 such inputs join the 11 tests in T^+, and the patch of Figure 2-(b) fails 10 of them. The search therefore continues and accepts the patch of Figure 2-(c), transplanted from a reference program with an explicit guard, which also passes the held-out tests of Section 5.6. The price is a larger patch, at RPS 0.84 against 0.47.

Escalation and learning. When both local strategies abstain, or the local patch is at least as large as the program it repairs, CoEP issues one query that attaches the minimum-edit reference program; on our benchmark, this happened for 12.5% of the submissions (Section 5.3). The Learn stage generalizes such repairs into new rules, which the tool keeps only if they reproduce the repair. Whichever path a repair takes, it passes T^+ (Section 4.7).

3 Preliminaries

3.1 Program Repair for Programming Assignments

The main aim of program repair for programming assignments is to turn each buggy submission into a fixed program that passes the assignment's test cases. To do so, a repairer iteratively generates, validates, and selects candidate programs for every submission. We define an assignment as a quadruple (d, T, B, R), where d represents the natural-language problem description, $T = \{t_1, \ldots, t_n\}$ denotes the instructor's test cases, B indicates the buggy programs, and R the reference programs.

Algorithm 1 Repairing an Assignment

Input: Description (d), tests (T), buggy programs (B), reference programs (R), local repair tool (L).
Output: Fixed programs (F).

```
1:  procedure RepairAssignment(d, T, B, R, L)
2:      F ← ∅
3:      for each b ∈ B do
4:          f ← fail
5:          if L ≠ ∅ then
6:              f ← LocalRepair(L, b, R, T)
7:          if f = fail then
8:              for each c ∈ Generate(d, T, b, R) do
9:                  if c ⊨ T then f ← c; break
10:         F ← F ∪ {(b, f)}
11:     return F
```

Writing $p \models T$ when a program p passes every test case in T, the submissions divide into $B = \{p \mid p \not\models T\}$ and $R = \{p \mid p \models T\}$. To repair $b \in B$ is to produce a fixed program f with $f \models T$; a candidate that passes only some test cases is not a repair.

Algorithm 1 describes a generic algorithm that repairs an assignment. It takes five inputs: the description (d), the test cases (T), the buggy programs (B), the reference programs (R), and a local repair tool (L). For now, let us set aside the last input, L, and the associated function, LocalRepair; their roles will be explained in Section 4. Existing techniques run Algorithm 1 without a tool (i.e., $L = \emptyset$). At line 2, the algorithm initializes the set F of fixed programs as an empty set. It then repeats the loop at lines 3–10 for each buggy program b. At line 8, the Generate function produces candidate programs from b, and line 9 keeps a candidate c that passes every test case; techniques differ in which passing candidate they keep (e.g., the one closest to b [44]). Finally, Algorithm 1 returns the set F of pairs of a buggy program and its fixed program, where $f = fail$ marks a program that was not repaired.

Existing techniques for programming assignments differ in how they implement the Generate function at line 8. Reference-guided techniques align b with a structurally matching reference program $r \in R$ and transplant the regions in which they differ: CLARA [9] and Sarfgen [36] align b with its nearest reference program, Verifix [1] backs the alignment with formal verification, and Refactory [11] refactors the reference programs to widen the range of alignable programs. They run offline at no marginal cost, but Generate returns no candidate when R holds no program whose structure matches b. LLM-based techniques implement Generate as a model query: PyDex [44] prompts the model with d, b, and three (buggy, correct) exemplars retrieved from the class and samples ten candidates, while others add conversation [38], tutoring [40], fine-tuning [14, 41], or fault localization [22, 23]. These techniques repair submissions that no reference matches, but each iteration of the loop at line 3 issues its own query, so the number of queries grows with $|B|$.

3.2 Limitation of Per-Submission Repair

Our key observation is that many submissions to one assignment need the same repair, yet Algorithm 1 derives each of them independently. In this work, we newly define a *recurring repair* as a repair whose edit another buggy submission to the same assignment also needs. More precisely, let $\pi(b)$ be the reference program that takes the fewest AST edit operations from b, and $\varepsilon(b)$ the multiset of edit operations that takes b to $\pi(b)$ (both defined in Section 4.3); the repair of b is recurring if $\varepsilon(b) = \varepsilon(b')$ for some other $b' \in B$.

Table 1 shows that 1,087 of the 2,970 buggy programs (36.6%) across the 11 assignments of our benchmark (Section 5.1) need a recurring repair, and 49.9% on the five first-year assignments. Nevertheless, the per-submission repairer PyDex [44] spent $292.77 on `claude-sonnet-5` querying for each of them, so that a single misconception is paid for once per student who makes it. Surprisingly, the share of recurring repairs falls to 2.7% on Codeforces problems rated 1200 to 2000, precisely where the reference-guided Refactory [11] repairs only 54.5% of the programs. This might stem

Table 1. Recurring repairs and the cost of repairing them one submission at a time. Distinct: distinct edit scripts to the minimum-edit reference program. Recurring: buggy programs whose edit script another buggy program of the same assignment also needs.

Corpus	Assignments	Buggy	Edit scripts			Refactory	PyDex
			Distinct	Recurring		RR (%)	Spend ($)
CS1	R1–R5	1,783	1,183	890	(49.9%)	90.8	146.27
Codeforces 800–1000	C1–C2	1,000	861	192	(19.2%)	83.1	113.71
Codeforces 1200–2000	C3–C6	187	184	5	(2.7%)	54.5	32.79
Total	11	2,970	2,228	1,087	(36.6%)	85.9	292.77

from the larger number of solution strategies that harder problems admit: each buggy program needs an edit of its own, and the pool seldom holds a program of the same structure.

One alternative is to make each query cheaper. However, this naive approach proved unsatisfactory. Prompt caching [2, 20], which bills a repeated prompt prefix at a tenth of the input rate, reduced PyDex's spend by only 10.2% on `claude-sonnet-5` and 17.3% on `gpt-5.4-mini`, since output tokens account for 76.6% of the former's spend and the variable buggy program precedes any shared content. Memoizing patches by submission text helps even less: only 8.6% of the buggy programs are textual duplicates of another.

3.3 Goal

The goal of this paper is to repair every buggy program of an assignment while spending model queries per assignment rather than per submission. Formally, we aim to synthesize a local repair tool L, with a number of builder queries that does not grow with $|B|$, such that the set

$$\{\, b \in B \mid \textsc{LocalRepair}(L, b, R, T) \neq \mathit{fail} \,\}$$

is maximized, where LocalRepair is the hook at line 6 of Algorithm 1; the remaining programs fall through to Generate at line 8. However, this is far from straightforward in practice. Across our benchmark, the 2,970 buggy programs need 2,228 distinct edit scripts, and even among recurring repairs, the largest group sharing one edit script holds only 48 programs (R4). These observations motivate our approach. Our work thus tackles two key challenges: (1) selecting, from thousands of distinct edits, the worked examples from which a builder can generalize a repair beyond the program it saw, and (2) trusting the code a builder writes without trusting its claims about that code.

4 Our Approach

4.1 Architecture

Figure 3 illustrates how CoEP synthesizes and deploys an assignment-specific repair tool while interacting with an LLM. The first two stages run once per assignment and build the tool. Select pairs each buggy program with its minimum-edit reference program and chooses one worked pair per cluster of edit scripts, without querying a model. In Synthesize, a builder LLM turns the worked pairs into a Python module of AST rewrite rules, each named after the misconception it repairs, and CoEP executes the test cases to return the pairs on which the rules still fail, up to a round budget. The builder also writes a generator of test inputs, which CoEP labels by the majority output of the reference programs to amplify T into T^+. The last two stages run on every submission. In Repair, the frozen tool tries its rules and then a hunk-level transplant, accepting a candidate only if it passes T^+, and issues a per-submission query only when both abstain or the local patch is larger

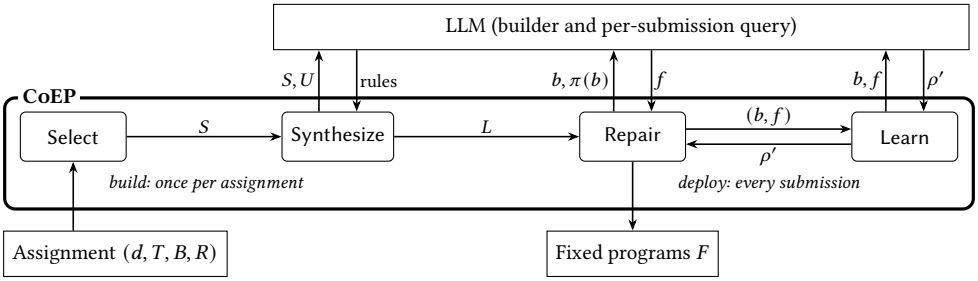


Fig. 3. Overview of CoEP

Algorithm 2 CoEP

Input: Description (d), tests (T), buggy programs (B), reference programs (R), builder LLM (M).
Output: Fixed programs (F), repair tool (L').

1: **procedure** CoEP(d, T, B, R, M)
2: /* **Step 1. Select** */
3: $S \leftarrow \text{Select}(B, R)$ ▷ $(b^\star, \pi(b^\star)) \in S$
4: /* **Step 2. Synthesize** */
5: $\langle U, j \rangle \leftarrow \langle S, 0 \rangle$ ▷ $\eta_{round} = 3$
6: **repeat**
7: $j \leftarrow j + 1$
8: $L_j \leftarrow \text{Synthesize}(M, d, T, S, U)$
9: $U \leftarrow \{(b, _) \in S \mid \text{Search}(L_j, b, T) = \mathit{fail}\}$
10: **until** $U = \emptyset$ **or** $j = \eta_{round}$
11: $L \leftarrow \arg\max_{i \le j} \text{cov}(L_i)$
12: $T^+ \leftarrow T \cup \text{Amplify}(M, d, T, R)$
13: /* **Step 3. Repair** */
14: $F \leftarrow \textsc{RepairAssignment}(d, T^+, B, R, L)$
15: /* **Step 4. Learn** */
16: $L' \leftarrow L$
17: **for each** $(b, f) \in F$ repaired by GENERATE **do**
18: $\rho' \leftarrow \text{Learn}(M, b, f)$
19: **if** $\text{Search}(L \cup \{\rho'\}, b, T^+) \neq \mathit{fail}$ **then** $L' \leftarrow L' \cup \{\rho'\}$
20: **return** F, L'

than the program it repairs. In Learn, the model generalizes each repair obtained from such a query into a new named rule, which the tool keeps only if the rule reproduces the repair.

The key novelty of CoEP lies in moving the model from the repair of each submission to the construction of the tool, which is ordinary code that runs offline and can be audited. Querying a model for every submission instead, as PyDex does, spends 21.3 times more for a comparable repair rate (Section 5.3).

4.2 CoEP

Algorithm 2 details how CoEP synthesizes a repair tool for an assignment and repairs its submissions. CoEP accepts the assignment (d, T, B, R) and a builder LLM (M), and returns the fixed programs (F) together with the tool. At line 3, the Select function chooses the build set S (Section 4.3). At lines 5–11, CoEP synthesizes the tool in at most η_{round} rounds: each round asks the builder for a module L_j (line 8), collects the worked pairs U it still fails to repair (line 9) for the next round, and the round with the highest coverage is kept (line 11). At line 12, the Amplify function builds the acceptance suite T^+ (Section 4.4). At line 14, CoEP repairs every buggy program by calling Algorithm 1 with the frozen tool L and T^+, and at lines 16–19, it learns new rules from the repairs obtained from a model query. In our experiments, η_{round} was set to 3, and every other constant below is fixed across all assignments.

4.3 Select

The goal of the Select stage is to choose a set of worked pairs from which the builder can generalize the recurring repairs of the class (line 3). This stage is local and issues no model query. To pair

programs, CoEP compares their ASTs. Let $\sigma(p)$ be the sequence of node types of p's AST read in breadth-first order, split into one sequence $\sigma_i(p)$ per top-level statement. With lev the Levenshtein distance between two such sequences, we define the distance between a buggy program b and a reference program r as the cost of the cheapest one-to-one matching $\mathcal{M}$ between their sequences:

$$\delta(b,r) = \min_{\mathcal{M}} \Big[\sum_{(i,j)\in\mathcal{M}} \mathrm{lev}(\sigma_i(b),\sigma_j(r)) + \sum_{i\notin\mathcal{M}} |\sigma_i(b)| + \sum_{j\notin\mathcal{M}} |\sigma_j(r)| \Big]$$

where an unmatched sequence costs its full length, so a missing or surplus definition costs what it is. Each buggy program then takes as partner the reference program it is fewest edits from, $\pi(b) = \arg\min_{r\in R} \delta(b,r)$, and $\varepsilon(b)$ denotes the multiset of edit operations that takes b to $\pi(b)$. Intuitively, $\pi(b)$ is the reference program that solves the problem the way b tries to, rather than the one that merely looks closest. When the node-type alignment yields no edit, $\varepsilon(b)$ is refined to an alignment over identifiers and literals, since a wrong variable or constant is invisible to node types alone.

CoEP then partitions B into clusters $C_1 \uplus \cdots \uplus C_k$ by average-linkage agglomerative clustering over the cosine distance between TF-IDF vectors of the edit scripts $\varepsilon(b)$, choosing k to maximize the mean silhouette coefficient [29]. We selected edit scripts as the clustering feature because a rule written from one member pays off only on the members needing the same edit; clustering on the programs' own AST paths instead filled the largest clusters with programs that merely look alike. SemCluster [25] clusters programs by semantic features because syntax over-partitions them, but the argument does not carry over to edits: an edit script cancels the syntax b shares with $\pi(b)$, and the consumer of a cluster, a rewrite rule, matches syntax. On R1, for instance, `gpt-5.4-mini`'s rule `strict_lt_to_lte` repairs both a two-line loop over `enumerate(seq)` and a four-way branch around a loop over `range(len(seq)-1)`; what the two share is the edit, not the program. Each cluster contributes its modal member,

$$b_i^\star = \arg\max_{b\in C_i} \big|\{b' \in C_i \mid \varepsilon(b') = \varepsilon(b)\}\big|,$$

whose edit the cluster most agrees on. We also considered the medoid and the member with the smallest edit. However, the medoid is usually a tangle of several faults, and the smallest edit belongs to an outlier that came closest to correct; the medoid serves only as the fallback for a cluster whose edits never repeat. The build set is $S = \{(b_i^\star, \pi(b_i^\star))\}_{i=1}^{k}$, ordered by increasing δ, so a weak builder is shown each buggy program with a reference program that repairs it and need only generalize the transformation. Empirically, Select chose 1,524 worked pairs over our 11 assignments, from 2 to 353 per assignment.

4.4 Synthesize

The goal of the Synthesize stage is to turn the build set into an assignment-specific repair tool and the suite that accepts its repairs (lines 5–12). At line 8, the Synthesize function prompts the builder M with the description d, up to $\eta_{test} = 8$ test cases of T as examples, and the worked pairs of S. The builder answers with a Python module L_j of AST rewrite rules, each named after the misconception it targets and tagged with the failure symptoms it addresses (e.g., a wrong answer or a `TypeError`). Formally, a rule is a candidate generator ρ that maps a program p to a set $\rho(p)$ of candidate programs. Figure 1-(b) shows one such rule. The contract on a rule is strict. A rule locates its edit on the AST and splices it in as a source-span replacement rather than re-rendering the module, which would reformat untouched lines; it never rewrites a whole function; and it proposes candidates rather than claiming a fix. A rule that cannot express a targeted edit, raises an exception,

or runs beyond an η_{alarm} = 2-second alarm yields $\rho(p) = \emptyset$, so its submission falls through to the next strategy.

Since a module written in one shot can contain its own defects, CoEP closes the loop on the build set itself, in the manner of counterexample-guided inductive synthesis [33] and of self-refinement from execution feedback [16, 30]: the worked pairs the rules still fail are the counterexamples. We define the coverage of a module as the fraction of the build set whose buggy half its rules repair:

$$\mathrm{cov}(L_j) = \frac{|\{(b, _) \in S \mid \mathsf{Search}(L_j, b, T) \neq \mathit{fail}\}|}{|S|}$$

where Search is the rule search of Section 4.5. Coverage is measured by executing T, never from the module's own claim, and it excludes the reference transplant, so a round is scored on the rules the builder wrote rather than on the substrate beneath them. While $\mathrm{cov}(L_j) < 1$, line 9 returns the still-failing pairs U to the builder for revision. The frozen tool L is the round with the highest coverage (line 11), ties going to the earliest. Intuitively, the refinement lets a builder repair its own rules from concrete failures, while the arg max ensures that a revision never replaces a better module.

Amplified acceptance. The instructor's test cases are few, 7.4 per assignment on R1–R5, and every strategy of the Repair stage accepts any candidate $c \models T$, so a thin T admits patches that fit the tests rather than the problem. At line 12, the Amplify function therefore enlarges the acceptance suite once per assignment, by differential testing against the reference programs. The builder M is prompted with d, T, and the instructor's reference solution, and writes a generator of random valid inputs, which CoEP validates as it validates rules: a round fails if the generator raises, returns a malformed input, or yields inputs of which the reference solution rejects more than 10%, and the failure is returned to the builder for at most η_{round} rounds. CoEP then draws η_{draw} = 300 fresh inputs and labels each with the majority output of η_{vote} = 40 programs sampled from R together with the reference solution, keeping an input only if at least 90% of them agree, so an input outside the specification, on which correct programs disagree, never becomes a test. The first η_{amp} = 100 such inputs form T_{amp}, and $T^+ = T \cup T_{amp}$. T^+ replaces T in every acceptance check of the Repair and Learn stages, whereas the coverage of the rules is still measured on T. This is differential testing [17] with the class as the oracle, in the spirit of test generation against overfitted patches [32, 39]; it adds one builder query per assignment and none per submission.

4.5 Repair

The goal of the Repair stage is to repair each buggy program b with the frozen tool, and to spend a model query only where the tool cannot (line 14). CoEP implements the LocalRepair hook of Algorithm 1 (line 6) with two strategies in increasing order of cost, and the Generate function (line 8) with a conventional model query; whichever first yields a candidate c with $c \models T^+$ supplies f.

Rule-based repair. The Search function composes the rules by a breadth-first search over layers of rewritten programs, $\Lambda_0 = \{b\}$ and $\Lambda_{j+1} = (\bigcup_{p \in \Lambda_j} \bigcup_{\rho \in L} \rho(p)) \setminus \bigcup_{i \leq j} \Lambda_i$, where the failing-test signature of b orders the rules whose declared symptoms match it. Every candidate in Λ_{j+1} is validated before Λ_{j+2} is enumerated, so the returned repair is the shallowest candidate c with $c \models T^+$. The search enumerates no further layer once η_{search} = 3 seconds have elapsed. Alongside the builder's rules, every tool carries a handful of always-on rules for the slips any class makes, such as flipped comparisons and off-by-one range and index shifts, which invert classical mutation operators [12, 13, 15]. Every repair the rules produce carries the ordered names of the rules behind it.

Reference-based repair. Most student bugs need a structural edit that no token-level rule expresses, and R supplies it. For each of the $\eta_{ref} = 8$ reference programs nearest to b under δ, CoEP computes the set $H(b, r)$ of line hunks in which b and r differ, both for r as written and for r with its variables renamed to b's (parameters by position, locals by first binding). Let $b \oplus h$ denote b with the hunks $h \subseteq H(b, r)$ applied. CoEP validates the $\eta_{cand} = 300$ subsets h with $1 \le |h| \le 3$ that change the fewest lines, and returns $\arg\min_{h \in P} \mathrm{RPS}(b, b \oplus h)$ over the first $\eta_{pass} = 5$ passing subsets P, abstaining when none passes. Here, the relative patch size $\mathrm{RPS}(b, f)$ is the tree-edit distance [45] between the ASTs of b and f divided by the size of b's AST [11]. This is reference reuse in the lineage of CLARA, Sarfgen, and Refactory [9, 11, 36], but at the granularity of line hunks rather than whole blocks: a hunk remains applicable when the student's strategy differs from the reference program's elsewhere.

LLM-based repair. When both local strategies abstain, CoEP attaches the minimum-edit reference program from the Select stage and issues a single query $\mathrm{LLM}(d, T, b, \pi(b))$, so that the model repairs b toward the student's own strategy. CoEP also escalates when the local strategies succeed with $\mathrm{RPS}(b, f_{loc}) > \eta_{rps} = 1$, i.e., with a passing patch at least as large as the program it repairs. The reason is that the acceptance test, $c \models T^+$, says nothing about patch size: a transplant can pass T^+ by replacing most of a student's program, whereas a rule must abstain whenever it cannot express a targeted edit. CoEP keeps

$$\arg\min_{f \in \{f_{loc},\, \mathrm{LLM}(d,T,b,\pi(b))\},\ f \models T^+} \mathrm{RPS}(b, f),$$

so an escalation can cost a query but never coverage or patch size. The threshold is a definition rather than a fitted value.

4.6 Learn

The goal of the Learn stage is to absorb the repairs the tool could not make into the tool itself (lines 16–19). For each pair (b, f) that Algorithm 1 repaired with a model query, the Learn function asks M to generalize the edit from b to f into a new named rule ρ'. CoEP keeps ρ' only if the tool extended with it repairs b by itself, i.e., $\mathrm{Search}(L \cup \{\rho'\}, b, T^+) \neq \mathit{fail}$. Learned rules are written to a staging copy L' that the current pass never consults, so no submission's outcome depends on which submissions preceded it, and L' replaces L only when the tool is frozen again. Since Learn queries the model only for submissions that already escalated, it adds no query for a submission the tool repairs.

The deployed tool itself never queries a model. Its local strategies run on a CPU in about three seconds per submission (Section 5.3), so student code leaves the institution only on escalation, and a defective rule degrades into abstention rather than a failing repair, because every candidate is validated against T^+ before it is returned.

4.7 Properties

We now state what CoEP guarantees whatever the builder writes. Throughout, $c \models T$ means that the candidate c passes every test case of T within its time limit, so a candidate that crashes or times out does not pass; $\mathrm{CoEP}_{\mathrm{L}}$ runs Algorithm 1 without Generate, and $\mathrm{CoEP}_{\mathrm{LF}}$ with Generate and the gate of Section 4.5.

Proposition 4.1 (Soundness relative to T^+). *For every builder M and every $b \in B$, if CoEP returns $f \neq \mathit{fail}$ for b, then $f \models T^+$, and hence $f \models T$.*

Proof. A repair enters F through one of three paths, and each validates its candidate against T^+ before returning it. Search returns a candidate only after validating it against T^+, and the transplant

returns the arg min over the passing subsets P; a candidate of Generate is kept only if $c \models T^+$ (line 9 of Algorithm 1, called with T^+ for T); and the gate keeps an arg min over candidates $f \models T^+$. Learned rules enter only through Search. The builder's rules and generator, the generated inputs, and the model's answers only propose candidates and tests; none of them is returned without being executed against T^+. Finally, $f \models T^+$ implies $f \models T$ because $T \subseteq T^+$. □

Proposition 4.1 implies that amplification can only shrink the set of accepted candidates: a repair under T may become an abstention or a different repair under T^+, but no candidate that T rejects is accepted. Section 5.6 measures what this exchange is worth: 0.6 points of repair rate on T for 8.4 to 10.2 points on held-out tests.

Proposition 4.2 (Shallowest composition). *If* Search *returns a candidate $c \in \Lambda_j$ for b, then no candidate in $\Lambda_0 \cup \cdots \cup \Lambda_{j-1}$ passes T^+; that is, c composes the fewest rule applications among the passing candidates of the enumerated layers.*

Proof. Search validates every candidate of Λ_j before it enumerates Λ_{j+1}, and stops at the first passing candidate. If c is returned from Λ_j, the layers $\Lambda_0, \ldots, \Lambda_{j-1}$ were therefore validated completely and held no passing candidate. Since each layer excludes the programs of earlier layers, an element of Λ_i is a program first reached by exactly i rule applications. □

Shallowest is not smallest, since one rule application may be a large edit; the contract on rules (Section 4.4) narrows the gap, as a rule splices a targeted edit as a source-span replacement and never rewrites a whole function.

Proposition 4.3 (Monotone escalation). *For every b, if CoEP$_L$ repairs b with f_L, then CoEP$_{LF}$ repairs b with some f_{LF} such that* $\mathrm{RPS}(b, f_{LF}) \leq \mathrm{RPS}(b, f_L)$.

Proof. CoEP$_{\mathrm{LF}}$ computes the same local repair $f_{loc} = f_L$. If $\mathrm{RPS}(b, f_L) \leq \eta_{rps}$, it does not escalate and $f_{LF} = f_L$. Otherwise, f_{LF} is the arg min of $\mathrm{RPS}(b, \cdot)$ over the candidates in $\{f_L, \mathrm{LLM}(d, T, b, \pi(b))\}$ that pass T^+, a set that contains f_L by Proposition 4.1 and is therefore nonempty. □

Proposition 4.4 (Termination). *For every b, the Repair stage terminates.*

Proof. Each rule call is cut off after η_{alarm}, each validation runs under a per-test time limit, and the rule set and every layer are finite. Search enumerates no further layer once η_{search} has elapsed, so it stops after finitely many layers. The transplant validates at most η_{cand} subsets for each of the η_{ref} reference programs, and Generate issues one query. □

Because the search budget is wall-clock time, which layers are enumerated, and hence which repair is returned, can depend on machine load; in practice, re-running the frozen `claude-sonnet-5` tool on R5 reproduced all 108 of its patches.

What is not guaranteed. CoEP is not complete: the transplant tries at most three hunks over at most 300 subsets and the search stops at its budget, so a repairable program may fall through to escalation or remain unrepaired. Nor is a repair correct beyond T^+: Proposition 4.1 bounds what CoEP accepts by T^+, not by the specification, and Section 5.6 measures the gap on held-out tests.

5 Experiments

In this section, we evaluate CoEP to answer the following research questions:

(1) **Effectiveness:** How effectively does CoEP repair buggy programs compared to reference-guided and LLM-based repair, in terms of repair rate and patch size?
(2) **Efficiency:** How much model spend and turnaround does synthesizing a tool once per assignment save, compared to querying an LLM per submission?

(3) **Efficacy of Each Stage:** How do the Synthesize, Repair, and Learn stages contribute to the performance of CoEP?
(4) **Generality:** Does a tool built from a class's earlier submissions repair its later ones, and how does its build scale with the class?
(5) **Correctness beyond the Tests:** Do the repairs remain correct on test cases that no arm searched against?

We implemented CoEP in Python on top of the standard `ast` module, and each synthesized tool is emitted as an ordinary Python module that runs offline. Tree-edit distance is computed with the Zhang–Shasha algorithm [45] and clustering with scikit-learn. All local computation ran on a Mac Studio (Apple M1 Ultra, 20 cores, 64 GB RAM, macOS 26.7).

5.1 Experimental Settings

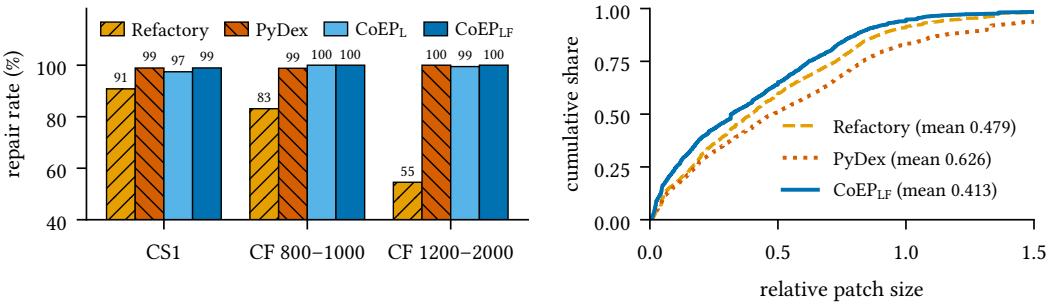
Baselines. We compared CoEP with two baselines representing the two mainstream lines of repair for programming assignments: (1) reference-guided repair (Refactory [11]) and (2) LLM-based repair (PyDex [44]); Section 6 explains why the other techniques could not serve as baselines. Refactory [11] refactors the class's reference programs to match each buggy program's structure and transplants the divergent blocks; it is the strongest reference-guided repairer that runs on our Python corpora. PyDex [44] prompts a model with each buggy program and the three most similar (buggy, correct) exemplars retrieved from the class, samples ten candidates, filters them with T, and returns the survivor closest to the student's program; we re-implemented its few-shot semantic phase and billed all ten samples. We also priced PyDex under prompt caching (PyDex_{PC}), modeling the first of its ten identical prompts at full price and the rest at the providers' cache-read rate [2, 20], which upper-bounds what caching can save on these queries. We evaluated CoEP in two configurations: CoEP_{L}, which runs the frozen tool alone with no model at repair time, and CoEP_{LF}, which adds the per-submission query on escalation. For the efficacy of each stage, we further evaluated CoEP_{L}–Syn, which omits the Synthesize stage (lines 4–11 of Algorithm 2) and keeps only the always-on rules and the reference transplant; CoEP_{L}+Learn, which re-runs CoEP_{L} with the learned rules merged and frozen beforehand; and $\text{LLM}_{\text{Direct}}$, which issues CoEP_{LF}'s query $\text{LLM}(d, T, b, \pi(b))$ for every buggy program. Since $\text{LLM}_{\text{Direct}}$ uses the same prompt, model, minimum-edit reference program, and acceptance test, it differs from CoEP_{LF} in one design choice alone: querying the model only where the frozen tool cannot serve.

Benchmarks. We used 2,970 buggy programs from 11 assignments in two corpora, summarized in Table 2. The first five assignments (R1–R5) are those of a first-year university programming course, with every submission its students wrote, released by the authors of Refactory [11]. The other six (C1–C6) are Codeforces problems whose submissions the open-r1 release mirrors [24]. So that difficulty extends beyond introductory exercises, we selected one problem at each of the ratings 800, 1000, 1200, 1400, 1600, and 2000, covering five official rating bands from Newbie to Candidate Master [5]. We kept each corpus's own labels, so the denominator of every arm is the same 2,970 programs.

Other Settings. Every LLM arm ran on two models, one per provider: `gpt-5.4-mini` ($0.75/$4.50 per million input/output tokens) and `claude-sonnet-5` ($2/$10). We selected each provider's cheapest tier at which prompt caching works on these prompts, to give the caching baseline its best case. Each model served both as CoEP's builder and fallback and as PyDex's repairer. On R1–R5, CoEP accepted repairs against the amplified suite T^+; on C1–C6, whose test cases are the judge's full official test sets (22 to 56 per problem), $T^+ = T$. We measured four metrics. The *repair rate* (RR) is the percentage of buggy programs an arm repairs, counting a repair only when the fixed program passes all test cases of T; on R1–R5, CoEP's acceptance is therefore stricter than the

Table 2. Benchmarks and repair results. R1–R5: a first-year course; C1–C6: Codeforces problems (rating). R1 Sequential search; R2 Unique dates and months; R3 Duplicate elimination; R4 Sorting tuples; R5 Top-k elements; C1 Nearly Lucky Number (800); C2 cAPS lOCK (1000); C3 Planning The Expedition (1200); C4 Measuring Lengths in Baden (1400); C5 Jumping Jack (1600); C6 Digital Root (2000). Buggy, Correct, Tests: programs and instructor's test cases per assignment. Bold: the best value in each row and metric, ties included.

ID	Buggy	Correct	Tests	Repair rate (%)				Relative patch size			
				Refactory	PyDex	CoEP$_{\text{L}}$	CoEP$_{\text{LF}}$	Refactory	PyDex	CoEP$_{\text{L}}$	CoEP$_{\text{LF}}$
R1	575	768	11	99.0	**99.7**	**99.7**	**99.7**	0.390	0.428	0.283	**0.276**
R2	435	291	9	78.2	**97.8**	91.1	97.1	0.560	0.428	0.427	**0.421**
R3	308	546	6	97.4	**99.8**	98.4	98.5	0.340	0.423	**0.297**	0.298
R4	357	419	6	88.2	99.3	**100.0**	**100.0**	0.310	0.535	0.387	**0.291**
R5	108	418	5	88.0	95.4	**100.0**	**100.0**	0.320	0.244	**0.205**	**0.205**
C1	440	353	34	85.2	99.9	**100.0**	**100.0**	0.588	0.685	0.584	**0.494**
C2	560	140	56	81.4	97.9	**100.0**	**100.0**	0.724	1.167	0.762	**0.648**
C3	69	164	31	47.8	**100.0**	98.6	**100.0**	0.847	0.705	0.638	**0.585**
C4	38	37	22	92.1	**100.0**	**100.0**	**100.0**	0.605	0.643	0.571	**0.525**
C5	77	2	30	40.3	**100.0**	**100.0**	**100.0**	0.878	0.914	0.914	**0.826**
C6	3	2	40	**100.0**	**100.0**	**100.0**	**100.0**	2.248	1.975	**1.652**	**1.652**
Total	2,970	3,140	250	85.9	98.9	98.4	**99.4**	0.500	0.634	0.481	**0.428**



(a) Repair rate by difficulty

(b) Patch size on the common pairs

Fig. 4. Repair rate and relative patch size of CoEP and the baselines.

baselines'. The *relative patch size* (RPS, Section 4.5) is averaged over the programs an arm repairs. The *turnaround* (ATT) is the mean wall-clock seconds per submission, split into local computation (ATT$_{cpu}$) and model latency (ATT$_{api}$). The *model cost* $C_{tot} = C_{build} + C_{api}$ is the US dollars spent over the benchmark, priced from exact token counts, where C_{build} is the one-off synthesis, amplification included, and C_{api} the per-submission queries. For Refactory on R1–R5, we report the repair rates and patch sizes its authors published, which leaves no room to argue that we misconfigured it; on C1–C6, we ran the released tool and measured RPS on the student's original program, as for every other arm. Effectiveness is pooled over the two builders (5,940 arm–submission pairs), and cost is reported per builder. We compared repair rates with the exact McNemar test on paired outcomes and patch sizes with the Wilcoxon signed-rank test, both at $p < 0.05$.

5.2 Effectiveness

Table 2 shows that CoEP_{LF} repaired as many programs as the strongest baseline while making the smallest patches. Overall, CoEP_{LF} repaired 99.4% of the 2,970 programs, against 85.9% for Refactory and 98.9% for PyDex, and reached 100% on eight of the eleven assignments. In terms of patch size, CoEP_{LF}'s mean RPS of 0.428 is 32.5% smaller than PyDex's 0.634 and also smaller than Refactory's 0.500. Since RPS is averaged over what each arm repairs, we also compared it on the programs both arms repaired: on the 5,841 pairs CoEP_{LF} and PyDex both repaired, their RPS was 0.426 against 0.634, and on the 5,023 pairs all three arms repaired, CoEP_{LF}'s 0.413 was below Refactory's 0.479 and PyDex's 0.626 (Figure 4-(b), where a curve further to the left means smaller patches).

Notably, the gap to Refactory widens with difficulty. Grouped into the CS1 assignments, Codeforces 800–1000, and Codeforces 1200–2000, Refactory's repair rate falls from 90.8% to 83.1% and 54.5%, whereas CoEP_{LF} holds 98.9%, 100.0%, and 100.0% (Figure 4-(a)). Thin pools alone do not explain this collapse, since C5 and C6 give every arm only two reference programs and both CoEP arms repair them completely. The difference lies in the granularity of reuse: Refactory adopts divergent blocks, so a pool that solves the problem another way leaves nothing to adopt, whereas CoEP adopts line hunks after renaming.

Interestingly, the frozen tool alone, CoEP_{L}, repaired 98.4%, slightly fewer than PyDex's 98.9%, without a single model call. Most of the gap lies on R2 and R3, where CoEP's acceptance is stricter: a candidate must also pass the 100 amplified test cases. The escalation of CoEP_{LF} recovered most of these programs, raising R2 to 97.1%, and Section 5.6 shows that the stricter acceptance is what keeps CoEP's repairs correct beyond T; the tool need not cover every misconception.

These results were statistically significant against Refactory (exact McNemar, $p < 0.001$ on both builders). Against PyDex, CoEP_{LF} repaired more programs with the builders pooled ($p = 0.014$) but not significantly on either builder alone ($p = 0.14$ and 0.065), and CoEP_{L} repaired slightly fewer when pooled ($p = 0.018$; $p = 0.15$ and 0.075 per builder). Both CoEP arms made smaller patches than both baselines on the common pairs (Wilcoxon signed-rank, $p < 0.001$). These results indicate that a tool synthesized once per assignment repairs as many programs as a model queried per submission, with patches a third smaller.

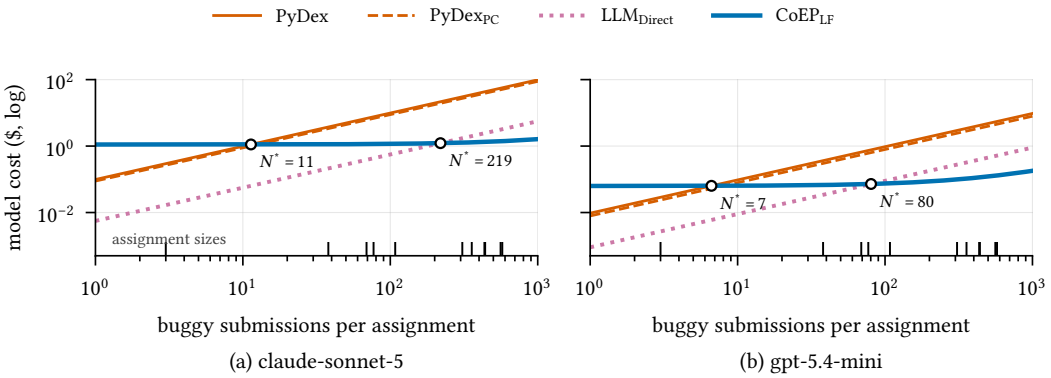
5.3 Efficiency

A per-submission arm pays for a query on every submission, whereas CoEP_{LF} pays its build once per assignment and queries only on escalation; the two break even at $N^* = \bar{C}_{build}/\Delta$ submissions per assignment, with $\bar{C}_{build}$ the mean build cost of an assignment and Δ the saving per submission. Figure 5 plots both costs on log scales, with circles at N^* and ticks at the sizes of our 11 assignments.

Table 3 shows that CoEP_{LF} spent \$13.74 over the benchmark on `claude-sonnet-5` and \$1.04 on `gpt-5.4-mini`, whereas PyDex spent \$292.77 and \$28.42, 21.3 and 27.3 times more. Prompt caching left PyDex_{PC} at \$263.05 and \$23.50, still 19.1 and 22.6 times more, because output tokens dominate the spend; even with every input token billed at the cache-read rate, whatever the order of the prompt, PyDex would cost \$231.14 and \$13.12, 16.8 and 12.6 times more. Notably, an assignment repaid its build after $N^* = 11$ submissions against PyDex on `claude-sonnet-5` and after 7 on `gpt-5.4-mini` (13 and 8 against PyDex_{PC}), and ten of the eleven assignments exceeded these points, most by an order of magnitude. Even against $\text{LLM}_{\text{Direct}}$, which sends a single query without exemplars, CoEP_{LF} cost 1.2 and 2.6 times less, although it breaks even later, at 219 and 80 submissions, and at that price $\text{LLM}_{\text{Direct}}$ repaired only 82.5% of the programs (Section 5.4). With the fallback, CoEP queried the model on 12.5% of the submissions (745 of 5,940 pairs), and those queries produced the repair on 8.5%.

Table 3. Model cost over the benchmark ($) and turnaround per submission (s). ×: the arm's C_{tot} over CoEP$_{LF}$'s. N^*: submissions per assignment at which CoEP$_{LF}$ breaks even with the arm. LLM$_{Direct}$'s turnaround is its model latency alone.

Arm	Model cost					Turnaround		
	C_{build}	C_{api}	C_{tot}	×	N^*	ATT$_{cpu}$	ATT$_{api}$	ATT$_{tot}$
`claude-sonnet-5`								
PyDex [44]	–	292.77	292.77	21.3	11	0.66	13.97	14.64
PyDex$_{PC}$	–	263.05	263.05	19.1	13	0.66	13.97	14.64
LLM$_{Direct}$	–	16.59	16.59	1.2	219	–	4.58	4.58
CoEP$_{LF}$	12.24	1.50	**13.74**	–	–	2.73	0.39	**3.12**
`gpt-5.4-mini`								
PyDex [44]	–	28.42	28.42	27.3	7	0.73	1.41	2.14
PyDex$_{PC}$	–	23.50	23.50	22.6	8	0.73	1.41	2.14
LLM$_{Direct}$	–	2.67	2.67	2.6	80	–	1.12	**1.12**
CoEP$_{LF}$	0.69	0.35	**1.04**	–	–	3.11	0.16	3.27



(a) claude-sonnet-5 (b) gpt-5.4-mini

Fig. 5. Model cost of one assignment against its number of buggy submissions.

Regarding turnaround, CoEP$_{LF}$ took 3.12 seconds per submission against PyDex's 14.64 on `claude-sonnet-5`, but 3.27 on `gpt-5.4-mini` against PyDex's 2.14 and LLM$_{Direct}$'s 1.12, since a per-submission arm's turnaround is almost all model latency and CoEP's almost all local search. What CoEP pays instead is a build of 22.2 minutes per assignment on `claude-sonnet-5` and 15.3 on `gpt-5.4-mini`, once before the first submission arrives.

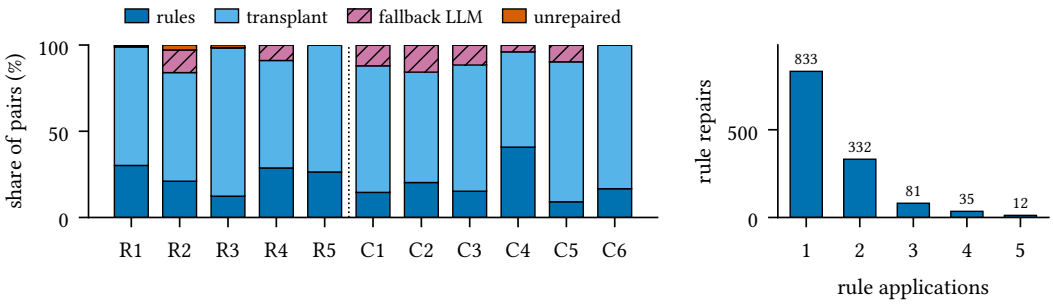
5.4 Efficacy of Each Stage

We evaluated the contribution of each stage with the variants of Section 5.1. Since the Select stage only feeds the Synthesize stage, we assess the two together by removing the builder's rules (CoEP$_{L}$−Syn).

Synthesize. Table 4 shows that the synthesized rules contribute coverage when the reference pool is thin, and smaller patches throughout. Over the full benchmark, CoEP$_{L}$−Syn repaired 98.25% at RPS 0.509, against CoEP$_{L}$'s 98.43% at 0.481; the difference of 11 pairs was significant on `claude-sonnet-5` (exact McNemar, $p = 0.004$) but not on `gpt-5.4-mini` ($p = 0.50$), and no program was repaired by CoEP$_{L}$−Syn alone. The gain in coverage is small because the reference

Table 4. Efficacy of the Synthesize stage. k: reference programs the transplant may use, drawn three times, on the 146 programs the builder never saw (`claude-sonnet-5`, mean). All (2,970 programs): the full benchmark, both builders.

Reference pool		$k=0$	$k=1$	$k=2$	$k=5$	All	All (2,970 programs)
Repair rate (%)	CoEP_L−Syn	8.2	97.3	98.4	99.5	99.3	98.25
	CoEP_L	**30.1**	**97.9**	**99.1**	99.5	99.3	**98.43**
Relative patch size	CoEP_L−Syn	0.103	1.128	0.925	0.688	0.494	0.509
	CoEP_L	0.208	**0.974**	**0.826**	**0.630**	**0.478**	**0.481**



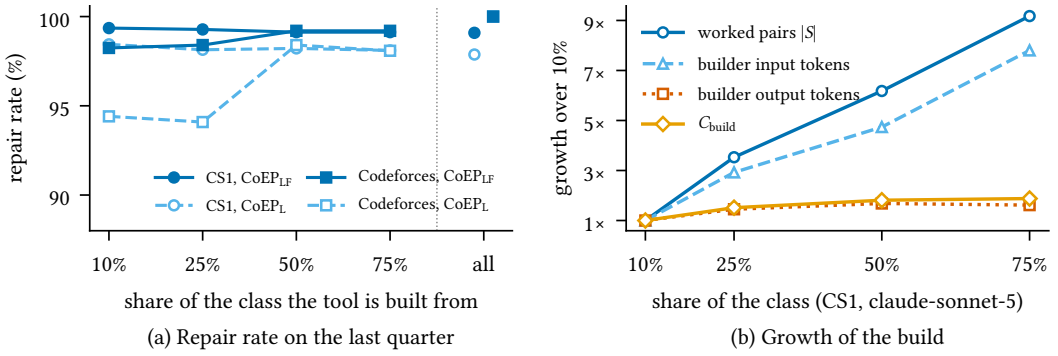
(a) Source of each repair (b) Composition depth

Fig. 6. Source of each repair and depth of the rule compositions.

transplant repairs nearly everything a rule repairs when a matching pool exists: of the 1,293 pairs the rules repaired, CoEP_L−Syn also repaired 99.1%. When the pool is thin, the rules matter: on programs the builder never saw, they raised the repair rate from 8.2% to 30.1% with no reference program, and cut RPS by 0.10 to 0.15 with one or two. The names also say what the rules do: three raters independently judged 191 rules against their code and a repair each produced (Fleiss' $\kappa = 0.890$), and 99.2% of the rule-produced repairs carry a name consistent with what the rule did; all 16 names judged wrong were written by `gpt-5.4-mini`, and 14 of them never fired.

Repair. Within the Repair stage, the reference transplant carried the breadth and the rules carried the precision. Of the 5,847 pairs CoEP_L repaired, the transplant produced 77.9% at mean RPS 0.587, and the rules 22.1% at 0.109, the smallest patches in the system; the rules' share ranged from 9.1% on C5 to 44.7% on C4 and was never zero (Figure 6-(a) gives the split under CoEP_LF). The rule repairs were not single rewrites: 460 of the 1,293 (35.6%) composed two to five rule applications (Figure 6-(b)), which the breadth-first search reached only after every shallower composition had failed (Proposition 4.2). The escalation trades a little spend for patch size: CoEP_LF raised the repair rate by 0.93 points and lowered RPS from 0.481 to 0.428.

Beyond a matching control flow. The transplant needs no correct program with the buggy program's control flow, the requirement CLARA calls the only fundamental problem of its approach [9] and Sarfgen imposes exactly [36]. Of the 5,902 pairs CoEP_LF repaired, 154 (2.6%) have a loop structure that no correct program of the class shares, leaving CLARA nothing to match, and 1,446 (24.5%) a nesting of loops and conditionals that none shares, leaving Sarfgen nothing to align; the tool's rules and transplant repaired 1,260 of these without a model query. Conversely, 2,171 of the 4,108 transplant repairs (52.8%) changed the buggy program's nesting of loops and conditionals.



(a) Repair rate on the last quarter

(b) Growth of the build

Fig. 7. Repair rate and build growth of tools built from the first f of a class.

Local-first architecture. The local-first architecture is what separates CoEP$_{LF}$ from LLM$_{Direct}$, which sends the same query for every program. CoEP$_{LF}$ repaired 99.4% against LLM$_{Direct}$'s 82.5% (exact McNemar, $p < 0.001$ on both builders), and LLM$_{Direct}$ repaired only 3 programs that CoEP$_{LF}$ did not: a single query has no second candidate when its patch fails, whereas the tool searches its rules and up to 300 hunk subsets before escalating. On the 4,896 pairs both repaired, however, LLM$_{Direct}$'s patches were smaller (RPS 0.330 against 0.391). It is therefore the minimum-edit reference program $\pi(b)$, which both arms attach, rather than the architecture that makes the patches small; the architecture contributes coverage and cost (Section 5.3).

Learn. The Learn stage fires only where the tool fails, which is rare. Over both builders, CoEP$_{L}$ failed on 93 pairs, on R1, R2, R3, and C3; the fallback repaired 55 of them, and the Learn stage generalized these repairs into 11 rules that passed the reproduction check, for \$0.89. CoEP$_{L}$+Learn recovered 32 programs with these rules without a single regression (exact McNemar, $p < 0.001$ and $p = 0.002$ for the two builders). The learned rules are evidence of reuse within an assignment, not of transfer across assignments. The 38 pairs CoEP$_{LF}$ left unrepaired all lie on R1–R3 and all reached the fallback: 25 are repaired by the same arm accepting against T alone, so only the amplified test cases rejected them, and for the other 13, all on R2, no candidate passed T itself.

5.5 Generality

We examined whether a tool built from a class's earlier submissions repairs its later ones. We ordered the submissions of C1–C5 by id, which follows submission time, and those of CS1, which records neither time nor student, in three seeded random orders (C6 has five submissions). We built a tool from the first $f \in \{10, 25, 50, 75\}\%$ of each order, its buggy programs feeding Select and its correct programs forming R, and repaired the buggy programs of the last quarter, 439 per order on CS1 and 313 on Codeforces, also with the tools of Section 5.2, built from the whole class, as a reference ("all" in Figure 7-(a)).

Figure 7-(a) shows that a tool built from the first 10% of a class repaired its later submissions about as well as one built from the whole class. On CS1, CoEP$_{LF}$ repaired 99.4% of the held-out programs from a 10% build, against 99.1% for the tool built from all submissions; on Codeforces, 98.2% against 100.0%, rising to 99.2% from half the class. The frozen tool alone needed more history on Codeforces: CoEP$_{L}$ repaired 94.4% from 10% and 98.4% from 50%, against 100.0% for the full-pool tool, since a thin pool offers fewer reference programs to transplant from, and the fallback recovered most of the difference. An instructor facing a new assignment can therefore build a tool from its first few dozen submissions and rebuild it as the term proceeds.

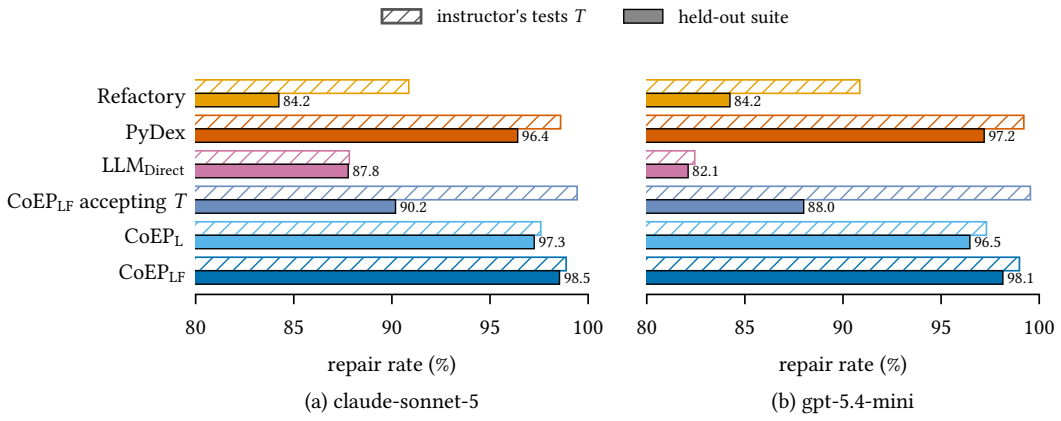


Fig. 8. Repair rates on R1–R5 under the instructor's test cases and a held-out suite. CoEP$_{\text{LF}}$ accepting T: CoEP$_{\text{LF}}$ accepting against T alone.

Notably, the build cost grew far more slowly than the class (Figure 7-(b)). From 10% to 75% of CS1, the build set grew 9.2-fold, from 15.6 to 143.1 worked pairs per assignment, whereas the build cost on `claude-sonnet-5` grew 1.9-fold, from \$0.88 to \$1.66 (`gpt-5.4-mini`: \$0.027 to \$0.069). The builder's input grew with the pairs, 7.8-fold, but its output, the module of rules that dominates the price, grew only 1.6-fold, since most new pairs fall into misconceptions the tool has already named.

CoEP also held across builders. The two builders differ 18-fold in build cost (\$12.24 against \$0.69), yet their tools repaired 98.52% and 98.35% of the programs, and CoEP$_{\text{LF}}$ repaired 99.33% and 99.39% at RPS 0.420 and 0.435, below PyDex's 0.598 and 0.669 on the same models. Coverage therefore does not hinge on a frontier builder, and a stronger builder pays off in patch size and in the misconception names.

5.6 Correctness beyond the Tests

R1–R5 ship 7.4 instructor test cases per assignment, and in Section 5.2 every arm both searched against them and was judged by them. To check whether the repairs remain correct beyond them, we built a held-out suite per assignment that no arm searched against: 100 inputs drawn with a different seed from a generator we wrote by hand, independently of the builders' generators, each labelled by the majority output of a different sample of 40 correct programs and the reference solution at 90% agreement, and none in T^+. We then re-ran every arm's repairs on T and the held-out suite; C1–C6 already use the judge's full official test sets.

Figure 8 shows that the amplified acceptance made CoEP's repairs the most robust of all arms. Of the programs CoEP$_{\text{LF}}$ repaired, 99.7% and 99.2% also passed the held-out suite, against 97.8% and 98.0% for PyDex, so on the held-out suite CoEP$_{\text{LF}}$ repaired 98.5% and 98.1% of the programs against PyDex's 96.4% and 97.2% (exact McNemar, $p < 0.001$ pooled; $p < 0.001$ and $p = 0.075$ per builder). Without amplification, the same tool accepted patches that fit T alone: only 90.7% and 88.4% of its repairs survived, and its held-out repair rate fell to 90.2% and 88.0%. Amplification thus traded 0.6 points of repair rate on T for 8.4 and 10.2 points on the held-out suite. Even the frozen tool, with no model call, reached 97.3% and 96.5% on the held-out suite, on par with PyDex.

Interestingly, the same amplification hurt the per-submission arm: LLM$_{\text{Direct}}$ repaired 87.8% and 82.1% on the held-out suite when accepting against T^+, but 90.2% and 83.8% against T alone. A single query has no second candidate, so a patch that departs from the class's behavior on inputs T

never exercises, such as sorted sequences with repeated elements on R1, is simply lost, whereas CoEP's rules and transplant search for another that passes. A stricter acceptance pays off only under a repairer that can search, and a tool built once per assignment can afford the search.

The remaining failures are few and explicable: the six on R5 all fail `top_k([], 0)`, the only empty input, which keeping the suites disjoint removed from T^+, and `gpt-5.4-mini`'s generator for R3 drew values only from −2 to 3, missing nine patches that fail on larger ones. These results indicate that CoEP's repairs are correct beyond the instructor's tests more often than per-submission repairs.

5.7 Threats to Validity

(1) **Class splits:** The CS1 corpus records neither time nor student, so its splits in Section 5.5 are random orders; a split by term may yield lower rates. (2) **Test adequacy:** A repair counts when it passes T. On R1–R5, CoEP also requires T^+ and Section 5.6 checks held-out tests, but both suites are labelled by the class's correct programs, so a behavior the whole class gets wrong goes undetected; on C1–C6, we rely on the judge's official tests. (3) **Builders:** Both builders are closed commodity models, so a rebuild may not reproduce the same rules; we release the 22 tools and 10 input generators, which run offline. (4) **Benchmarks:** Our 11 assignments are Python problems, and C6 contributes three buggy programs; the results may not carry over to other languages or to large projects.

6 Related Work

Program Repair for Programming Assignments. Our work shares a common goal with techniques that repair students' programming assignments [1, 4, 6, 7, 9, 11, 14, 22, 23, 36, 40, 41, 44]. Singh et al. [31] repair an attempt with an instructor-written *error model*, a set of rewrite rules whose combinations a Sketch-based search explores for a minimal correction; CoEP synthesizes that error model per assignment and searches it by enumeration validated against T^+. Reference-guided techniques reuse the class's correct programs: CLARA [9] and Sarfgen [36] align a buggy program with a correct one of the same loop structure or the same sequence of control statements, which a quarter of our repairs lack (Section 5.4), Verifix [1] backs the alignment with formal verification, Refactory [11] refactors the pool to widen what aligns, and MENTORED [4] recombines correct and incorrect programs by genetic programming. LLM-based techniques query a model per submission, with retrieved exemplars (PyDex [44]), tutoring conversation (CREF [40]), fine-tuning (FastFixer [14], MORepair [41]), preference-trained localization (AdaPatcher [6]), MaxSAT-based localization with counterexample-guided synthesis [22, 23], or retrieval of other students' (incorrect, corrected) pairs (LSGen [7]). Unlike these techniques, CoEP spends its model effort on a tool built once per assignment and leaves no model in the deployed repairer; any of them can serve as the GENERATE function behind its tool.

Clustering and Transforming Submissions. OverCode [8] groups correct programs whose normalized statements coincide, CLARA [9] those with the same control flow and variable traces, SemCluster [25] those with similar per-path input counts and value-change frequencies, and InvAASTCluster [21] those with matching invariants and anonymized ASTs, each to choose the reference program a repairer aligns with. CoEP instead clusters buggy programs by the edit that repairs them, since its clusters decide which repairs a builder generalizes. Head et al. [10] group incorrect submissions by the transformations that fix them, so that a teacher writes feedback once per group. Like CoEP, Refazer [27] induces assignment-specific transformations from (before, after) pairs, but synthesizes them by programming-by-example in a domain-specific language from students' own fixes, whereas CoEP has an LLM write named, test-validated rules as general-purpose code. Template-based repair [13, 15] and Reyes et al. [26] apply mined or LLM-generated AST transformations to software projects rather than to a class.

LLMs as Tool Makers. Our approach belongs to a line of research in which LLMs write reusable tools instead of solving each task directly [3, 19, 28, 34, 35, 37, 43]: LATM [3] has a strong model write utilities that a weaker one reuses, CRAFT [43] creates and retrieves task-specialized toolsets, TroVE [37] and ReGAL [34] induce verifiable function libraries, Voyager [35] accumulates skills, FunSearch [28] evolves programs against a score, and ReaComp [19] compiles model reasoning into symbolic solvers. However, the first five keep a model in the loop that calls the tools at inference, whereas CoEP deploys its tool with no model behind it. What program repair adds is an executable oracle for the tool itself. Closer to programming languages, type-constrained decoding guarantees well-typed LLM output by checking each prefix against the type system [18]; CoEP obtains its guarantee instead by executing everything the LLM writes against T^+ (Proposition 4.1).

Selection of Baselines. We selected the baselines of Section 5.1 by four criteria: a technique must (1) produce repairs rather than clusters or tools for other tasks, (2) repair Python programs, (3) require no input our benchmark lacks, such as the link between a student's successive attempts, a tutor in the loop, or a model trained for the task, and (4) represent its line of repair on the CS1 corpus. Only Refactory and PyDex satisfy all four. Clustering, teacher-in-the-loop feedback, and tool-making approaches [3, 8, 10, 21, 25, 34, 37, 43] fail (1); Verifix [1], SemCluster, InvAASTCluster, and the counterexample-guided repairers [22, 23] (C), Sarfgen [36] (C#), and template- and transformation-based repair [13, 15, 26] (software projects) fail (2); and Refazer [27] and LSGen [7], which learn from or retrieve a student's successive attempts, Singh et al. [31], which needs an instructor-written error model per assignment, CREF [40], which keeps a tutor in the loop, and FastFixer [14], AdaPatcher [6], and MORepair [41], which fine-tune a model, fail (3). Of the remaining techniques, CLARA [9] and MENTORED [4] reuse the class's submissions as Refactory does, and we take Refactory, whose authors released the CS1 corpus and evaluated it against CLARA there [11], as the representative of that line; PyDex, unlike general-purpose repairers such as ChatRepair [38], is designed for introductory Python assignments and publishes its prompts in full [44].

7 Conclusion

Recently, LLM-based program repair has substantially improved the share of student submissions that can be repaired automatically. However, these techniques query the model once per buggy submission, so a class pays again for every recurring repair. This paper presents CoEP, which amortizes recurring repairs by having an LLM synthesize a repair tool once per assignment rather than a patch for every submission. CoEP selects worked pairs by clustering the class's edit scripts, synthesizes misconception-named rewrite rules, amplifies the test cases by differential testing against the class's correct programs, repairs every submission locally, and learns new rules from the few it escalates. Experimental results demonstrate that CoEP matches per-submission LLM repair in repair rate while producing smaller patches, repairs that hold on held-out tests more often, and a fraction of the model spend. We expect CoEP to make automated repair affordable for large classes, since a tool built from a class's first submissions repairs the rest.

Data-Availability Statement

CoEP's implementation, the benchmark with its held-out and amplified test suites, the synthesized tools, every prompt, and the per-submission logs, which reproduce every number in this paper without model access, are available at https://anonymous.4open.science/r/coep/.

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